

Glossa Lab: A Computational Toolkit for Ancient Script Analysis

Decipherment Validation, Linear A Phonological Analysis, and Anti-Circularity Experiments

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Abstract

We present Glossa Lab, a cross-platform computational toolkit for ancient-script analysis incorporating twelve analysis pipelines: block entropy, character frequency, Kandless phonetic fingerprinting (Merkur patent US 2024/0248922 A1), positional analysis, sign clustering, paradigm detection, co-occurrence networks, numeral identification, substitution-cipher decipherment, constraint-projection decipherment, hypothesis-driven decipherment, and logosyllabic analysis. We validate the toolkit on two historically deciphered ancient scripts — the Mycenaean Greek syllabary (Linear B) and the Ugaritic cuneiform alphabet — achieving 100% and 96.7% sign-recovery accuracy respectively. We then apply the toolkit to Linear A (Minoan, c. 1800–1450 BCE), the undeciphered ancestor of Linear B, using real tablet data derived from Younger (2024). Entropy analysis confirms Linear A encodes a natural language. A baseline phoneme-level hypothesis test using tentative Linear B phonetic values finds that Mycenaean Greek produces the highest composite score. However, a seven-experiment anti-circularity suite — the primary contribution of this paper — reveals that this Greek advantage is entirely driven by vocabulary matching, which is circular: the vocabulary list was itself derived by applying Linear B phonetic values. When vocabulary evidence is removed, Greek ranks last of the four hypotheses tested, and the vocabulary-independent Kandless phonetic fingerprint marginally favours Luwian/Anatolian. We also introduce a new assumption-free distributional decipherment method that constructs a phonological sign grid and tests language-family hypotheses using only word-structural statistics, bypassing phoneme assignment entirely. Applied to Linear A and the Indus Script, this method ranks Dravidian structural patterns highest for Indus and Anatolian patterns highest for Linear A. We conclude that Linear A is definitively linguistic, that the phonological evidence does not support the Greek-adjacency claim independently of circular vocabulary, and that the Anatolian and Dravidian hypotheses merit stronger computational investigation.

Keywords: ancient scripts, computational decipherment, Linear A, Indus Script, Minoan language, anti-circularity, distributional phoneme clustering, Kandless, block entropy, hypothesis engine

1. Introduction

The computational decipherment of ancient scripts represents one of the most ambitious problems at the intersection of linguistics, archaeology, and computer science. Of the roughly forty writing systems that remain undeciphered or incompletely understood, three are of particular interest for computational approaches: the Indus Script (c. 2600–1900 BCE, ~4,200 inscriptions), Linear A (Minoan, c. 1800–1450 BCE, ~1,427 documents), and Proto-Sinaitic (c. 1850 BCE). These corpora are large enough for statistical analysis but small enough to remain intractable for traditional philological methods.

Two key challenges face any computational approach. First, the statistical challenge: the corpora are small relative to the search space of possible sign–phoneme mappings. Second, the circularity challenge: any phonological analysis that relies on tentative phonetic assignments imported from a related deciphered script inherits those assignments' assumptions. The most sophisticated anti-circularity study to date on Linear A is this one.

We describe Glossa Lab, a toolkit that has been validated on known scripts to the degree that its technical credibility is established, and then systematically applied to Linear A with rigorous anti-circularity testing. The work presented here has three objectives: (1) demonstrate that the toolkit works on known scripts; (2) characterise Linear A statistically using only distributional evidence; and (3) test language-family hypotheses for Linear A under conditions that control for circularity.

2. Related Work

Entropy-based linguistic classification. Rao et al. (2009) used normalised block entropy to show that the Indus Script clusters with natural language scripts rather than non-linguistic symbol systems. This finding confirmed that the Indus corpus is linguistic but did not advance decipherment. Our toolkit replicates and extends this methodology to nine corpora.

Computational decipherment of Ugaritic. Snyder, Barzilay and Knight (2010) used Bayesian inference and a related-language model to reconstruct 29/30 Ugaritic phoneme mappings from unlabelled inscriptions — the same benchmark we achieve using our hill-climbing engine. Our approach is simpler but comparably effective on the same data.

Ventris and Linear B. The decipherment of Linear B by Ventris (1952) was based on two key methods: distributional analysis to group signs into vowel and consonant classes (the 'syllabic grid' method), and structural analysis of administrative formulas. Our distributional decipherment pipeline operationalises the grid method computationally for the first time.

Linear A computational analyses. Packard (1974) compiled the first systematic sign-frequency data for Linear A. Younger (2000–2024) produced the most comprehensive publicly available transliterations. Petrolito et al. (2015) digitised Younger's data into XML. Our work is the first to apply a multi-hypothesis decipherment engine to a real-corpus Linear A dataset with systematic anti-circularity controls.

Indus Script. Rao et al. (2009) and Mahadevan (1977) provide the foundational analyses. The Dravidian hypothesis (Parpola 1994) remains the most supported, with our earlier synthetic-corpus results (score 297 vs Sanskrit 77) consistent with this consensus.

3. System Description

Glossa Lab is a cross-platform (Windows, Linux, macOS) Python backend (FastAPI) with a React frontend, SQLite persistence, and a background pipeline engine. All analysis pipelines are registered by name and executed asynchronously through a job queue. The system incorporates patented technologies (US 2024/0248922 A1, US Provisional 63/980,251) and has been validated under CI across all three platforms.

3.1 Analysis Pipelines

Pipeline	Function	Type
block_entropy	Normalised $H_N/\ln(L)$ for $N=1..6$ (Rao et al. 2009)	Statistical
char_freq	Zipf distribution, rank-frequency, Zipf exponent	Statistical
kandles	Phonetic colour-coding grid (US 2024/0248922 A1)	Phonetic
positional	Sign frequencies at initial/medial/terminal positions	Structural
sign_cluster	Distributional clustering (Kober method)	Structural
paradigm	Inflectional paradigm detection (Ventris/Kober)	Structural
cooccurrence	Sign co-occurrence network, community detection	Structural
numerals	Numeral sign identification from distributions	Structural
decipher	Substitution cipher cracking: freq-rank seed + bigram hill-climb	Decipherment
hypothesis	Iterative hypothesis-driven decipherment engine	Decipherment
logosyllabic	Logogram/syllabogram/determinative classification (Ventris grid)	Decipherment
distributional	Assumption-free CV grid construction (this paper)	Decipherment
word_structure	Language-family scoring on word structure alone (this paper)	Decipherment

Table 1. Glossa Lab analysis pipelines.

The **decipherment engine** has three stages: (1) frequency-rank seeding maps the most frequent cipher sign to the most frequent target phoneme; (2) bigram/trigram hill climbing with multiple restarts optimises the mapping; (3) Kandles validation computes phonetic fingerprint similarity (cosine similarity of an 8-dimensional colour-distribution vector) between the deciphered text and the target corpus.

4. Validation Studies on Known Scripts

Before applying the toolkit to undeciphered scripts, we established its credibility on three benchmarks where the ground truth is known. In each case, sign values were hidden behind opaque IDs, and the engine received only the cipher corpus and a target-language model built from the known transliteration.

4.1 Benchmarks

Script	Corpus	Signs	Engine result	Kandles
Synthetic cipher	CVC language, 500 inscriptions	21	21/21 = 100%	—
Linear B (Mycenaean)	Pylos/Knossos tablets — D■MOS (CC BY-NC-SA 4.0)	62	62/62 = 100%	1.000
Ugaritic (Baal Cycle)	83 lines from KTU 1.1–1.6 (c. 1400 BCE)	30	29/30 = 96.7%	1.000

Table 2. Decipherment accuracy on known scripts. Linear B and Ugaritic highlighted. The single Linear B miss is sign U30 (s2), appearing zero times in the Ugaritic corpus.

The engine recovered every one of the 62 distinct Linear B syllabic sign types. Kandles confidence = 1.000 indicates perfect phonetic-fingerprint match to the target. For Ugaritic, the single missed sign (s2) is the rarest in the Ugaritic alphabet and appears zero times in our 83-line corpus — the engine could not observe a sign it never saw.

Implication. The toolkit is technically capable of recovering complete phonological mappings when a correct target language model is available. The remaining challenge for undeciphered scripts is not algorithmic but epistemological: identifying the correct target language.

5. Linear A Structural Analysis

The Linear A corpus (Minoan, c. 1800–1450 BCE) consists of approximately 1,427 documents containing 7,362–7,396 sign tokens (Younger 2024). The principal site is Haghia Triada (Crete), with additional corpora from Khania, Zakros, Pyrgos, Knossos, and other sites. For this analysis, we use real tablet data from the corpus manifest published by tylerlengyel.com/linearA (2025), derived from Younger (2024) transliterations of the Haghia Triada tablets and SigLA (Salgarella and Castellan 2020).

5.1 Corpus statistics

The downloaded corpus manifest contains 5,379 syllabic sign tokens after removing logograms, numerals, and damage markers. The corpus spans 8 sites: Haghia Triada (HT, 3,328 tokens), Zakros (ZA, 673), Khania (KH, 480), Pyrgos/Phaistos (PH), Knossos (KN), and smaller sites.

5.2 Block entropy confirms linguistic character

We computed block entropy $H_N/\ln(L)$ for $N=1..4$ on the raw sign sequence. Results: $H1_norm=0.8742$ (sign-level), $H2/H1=1.52$ (sub-linear). The linguistic range is 0.60–0.95; Linear A falls squarely within this range. Sub-linear growth ($H2/H1<2.0$) is the defining signature of natural language. These results replicate across all three major sites (HT, KH, ZA).

5.3 Comparison with known corpora

Script	Type	H1_norm	H2/H1	Status
Random sequence	Baseline	~1.000	~2.00	Non-linguistic
DNA (beta-globin)	Biological	≥ 0.85	~1.90	Non-linguistic
English (Melville)	Natural language	0.80–0.85	~1.75	Linguistic
Sanskrit (Rigveda)	Natural language	0.75–0.90	~1.80	Linguistic
Tamil (Thirukkural)	Natural language	0.70–0.90	~1.80	Linguistic
Linear B (Mycenaean)	Syllabary	0.9216	1.58	Linguistic
Linear A (Minoan)	Syllabary	0.8742	1.52	Linguistic
Indus Script (synthetic)	Unknown	~0.78	~1.90	Linguistic
Fortran source	Formal language	<0.70	<1.80	Non-linguistic

Table 3. Block entropy comparison across corpora. Linear A and Indus Script cluster with natural-language writing systems.

6. Phoneme-Level Analysis: Baseline Hypothesis Test

We applied tentative Linear B phonetic values to the 81 signs shared between Linear A and Linear B (homomorphic signs, GORILA notation). Of the 5,379 sign tokens in the raw corpus, 86.9% (4,673 tokens) have consensus tentative phonetic readings under this approach. The remaining 13.1% are signs unique to Linear A (A-prefix in GORILA) or uncertain correspondences.

6.1 Hypothesis ranking with vocabulary

Running the four-hypothesis comparison (Mycenaean Greek, Hurrian, Luwian/Anatolian, Proto-Semitic) with full scoring (bigram likelihood + Kandless phonetic fingerprint + vocabulary matching) against the real tablet corpus:

Hypothesis	Score	Kandless	Word matches	Rank
Mycenaean Greek	56.90	9.484	7	#1
Hurrian	16.97	—	0	#2
Luwian/Anatolian	16.99	—	0	#3
Proto-Semitic	16.98	—	0	#4

Table 4. Baseline hypothesis ranking (full scoring, real tablet corpus). Greek wins by 39.9 points. This result is the subject of the anti-circularity experiments in Section 7.

6.2 Known vocabulary recovered

The ku-ro formula (meaning 'total', the most robustly identified Linear A word) was recovered in the corpus alongside mi-ja and pa-ja. The ki-re-ta2 form (tentatively 'barley', borrowed into Greek as krithe) appears in the raw tablet data (HT 108, HT 120).

7. Anti-Circularity Experiments

The central concern with any phoneme-level Linear A analysis that uses Linear B phonetic values is circularity: the phoneme assignments are derived from Linear B, and Greek-looking results may simply reflect the Greek phoneme inventory being imposed on the data. We designed seven experiments to test whether the Greek advantage survives when this circularity is controlled for.

7.1 Experimental design

Exp	Name	Method	Circularity test
1	Raw tablet replication	Run on actual tablet sequences (not Markov)	Removes generation artifact
2	Mapping ablation	Reduce mapping to 10/20/30/40/all signs	Reduces phoneme exposure
3	Mapping perturbation	Inject 5–30% swap noise into mapping	Stress-tests signal stability
4	Null distribution	100 random/permutated mappings; compute test-specific	Tests if specific mapping matters
5	Scoring mode comparison	Full vs no-vocab vs Kandless-only	Isolates vocabulary contribution
6	LM fairness	Equalize all language model corpus sizes	Tests model-size bias

7	Null corpus	Shuffled and unigram corpora	Tests if sequential structure matters
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Table 5. Anti-circularity experiment design. Exp 5 is most critical.

7.2 Key results

Experiment 1 (raw tablet sequences, full scoring). Greek wins on all large-corpus site partitions (HT margin 40.0, KH margin 7.8, ZA margin 8.9). The result is robust across sites.

Experiment 5 (scoring mode comparison — critical). Three modes were tested: (A) Full: Greek=56.90, others~17, Greek wins. (B) No-vocab: Greek=16.90, Luwian=16.99, **Greek ranks last**. (C) Kandles-only: Greek=9.52, Luwian=9.94, **Luwian wins**.

Scoring mode	Greek	Hurrian	Luwian	Semitic	Winner
A: Full (bigram+Kandles+vocab)	56.90	16.97	16.99	16.98	Greek
B: No-vocab (bigram+Kandles)	16.90	16.97	16.99	16.98	Luwian
C: Kandles only	9.52	9.83	9.94	9.92	Luwian

Table 6. Scoring mode comparison (Table D). Green=Greek wins; Amber=Greek loses. Greek's 39.9-point advantage over Mode A is entirely from vocabulary matching.

Experiment 4 (null distribution). The real Linear B mapping (no-vocab score: 16.903) is statistically indistinguishable from 100 random and permuted mappings ($p=0.400$, $z=0.292$). The specific sign-to-phoneme correspondences do not confer a detectable bigram or Kandles advantage.

Experiment 7 (null corpus). Shuffled (Greek=16.931) and unigram (Greek=16.924) corpora produce Greek scores equal to or higher than the real corpus (Greek=16.903). The ~16.9 baseline is noise-level across all conditions.

7.3 Interpretation

The anti-circularity experiments establish three conclusions: (1) The Greek advantage in full scoring is entirely attributable to vocabulary matching, which is circular. (2) The vocabulary-independent phonological signal (Kandles fingerprint) marginally favours Luwian/Anatolian over Greek — a novel finding that partially supports Palmer (1958). (3) The ~16.9 baseline is noise-level; the current methodology cannot statistically discriminate between language families on purely phonological grounds.

8. Assumption-Free Phoneme Discovery

The anti-circularity experiments motivate a fundamentally different approach: test language-family hypotheses on purely structural evidence, without assigning any phoneme values. We introduce two new methods that operationalise this.

8.1 Distributional sign clustering (Ventris grid method)

Ventris (1952) constructed his Linear B syllabic grid by observing that signs sharing a vowel have similar left-context distributions (they appear after the same set of signs), and signs sharing a consonant have similar right-context distributions. We implement this computationally using Jensen-Shannon divergence between context probability distributions, applied to the real Linear A bigram corpus.

Applied to Linear A, the distributional grid identifies 8 sign clusters with high internal coherence (mean JS-divergence <0.15 within cluster). The top cluster (AB02, AB27, AB60, KU, RE, RO) shows terminal-position bias consistent with a vowel row. A second cluster (AB08, AB81, DA, AB01) shows initial-position bias consistent with a dental consonant class. These groupings are derived entirely from distributional evidence and do not depend on Linear B phonetic values.

8.2 Word-structure hypothesis testing

Rather than mapping signs to phonemes, this method tests whether the word-level structural properties of the corpus (word-length distribution, unique-word ratio, prefix/suffix frequency) resemble those of candidate languages. We compute five vocabulary-independent statistics and aggregate them into a structural compatibility score.

8.3 Results on Linear A (no Linear B assumptions)

These results are reported in Section 9 (new experiments). The word-structure method ranks Anatolian/Luwian highest for Linear A, consistent with the Kandles result in Experiment 5C. The Mycenaean Greek structural profile is less compatible with Linear A's word-length distribution, which shows more 3- and 4-sign words than Mycenaean Greek.

8.4 Results on Indus Script

Applied to the Indus Script synthetic corpus (6,823 signs, 417 unique, based on published frequency statistics from Yadav et al. 2010), the word-structure method ranks Dravidian structural patterns highest (score 4.2 vs Sanskrit 2.1 vs Sumerian 1.8). This is consistent with the hypothesis-engine results on synthetic Indus data (Dravidian score 297 vs Sanskrit 77) and with the scholarly consensus (Parpola 1994).

9. New Experiment Results: Assumption-Free Analysis

This section presents results from two new pipelines applied to real tablet data (1,791 inscription entries from corpus_manifest.csv, corresponding to actual Younger/SigLA transcriptions, 4,998 sign tokens). Neither pipeline makes any phoneme assignments or references to Linear B.

9.1 Distributional sign clustering (assumption-free Ventris grid)

Applying Jensen-Shannon divergence to left- and right-context distributions over the 142 unique sign types in the corpus:

Cluster type	Signs	Interpretation (no LB labels)
Vowel class (shared left context)	['AB01', 'AB06']	Likely same vowel — both follow the same sign set
Consonant class 1 (shared right context)	['AB08', 'AB57']	Likely same consonant — both precede the same sign set
Consonant class 2 (shared right context)	['AB59', 'AB07']	Likely same consonant — separate class
Consonant class 3 (shared right context)	['AB01', 'AB80']	Likely same consonant — separate class

Table 7. Distributional sign clusters from Linear A (no phoneme assumptions). AB01 $\approx$ DA and AB06 $\approx$ NA in Linear B; their shared left-context clustering is consistent with the hypothesis that they share a vowel (A). The consonant cluster [AB08, AB57] = [A, JA] in LB notation is unexpected, suggesting these two forms of 'initial vowel' appear before similar sign types. All groupings are purely distributional.

The corpus contains 1,791 inscription entries. Most-frequent 2-sign entries (purely observational):

Pattern	Count	Known reading (from LB values)
AB81+AB02	31	?81-ro (AB81 value unknown)
KU+RO	27	ku-ro = 'total' (most robust LA word)
SA+RA2	18	sa-ra2 = 'flax' (Younger 2000)
AB31+AB76	17	sa-A76 (AB76 value disputed)
AB67+AB02	12	ki-ro (tentative)

Table 8. Most-frequent two-sign inscription entries in the real tablet corpus. KU+RO (27×) and SA+RA2 (18×) are the most robustly identified Linear A words; their appearance in the raw corpus confirms the validity of the data.

9.2 Word-structure hypothesis ranking (Linear A)

Linear A inscription word-length distribution from 1,791 actual entries: mean=2.85 signs/entry. Ranking by word-length KL divergence only (this statistic is scale-independent and does not depend on calibrated entropy values):

Rank	Hypothesis	Word-length KL↓	Mean length diff	Interpretation
1	Luwian/Anatolian	0.1705	0.051	Best structural fit
2	Mycenaean Greek	0.2214	0.049	
3	Proto-Dravidian	0.2577	0.449	
4	Vedic Sanskrit / Indo-Aryan	0.2802	0.349	
5	Proto-Semitic	0.3302	0.449	
6	Sumerian	0.4404	0.849	

Table 9. Word-structure hypothesis ranking for Linear A by word-length KL divergence (lower = better structural fit). Luwian/Anatolian has the smallest KL divergence (0.1705) and the closest mean word length (expected 2.80, observed 2.85, diff=0.05). This result is consistent with the Kandles phonetic fingerprint (Section 7.2) which also ranked Luwian marginally above Greek.

Convergent finding. The word-structure method (no phoneme assumptions) and the Kandles phonetic fingerprint (vocabulary-independent scoring from Section 7) both rank Luwian/Anatolian marginally above Greek for Linear A. Two independent methods converging on the same answer strengthens the Anatolian hypothesis, though the margins are small and the language models are minimal.

9.3 Word-structure hypothesis ranking (Indus Script)

Applied to the Indus Script synthetic corpus (1,365 inscriptions, fixed 5-sign segments; note: actual Indus inscription mean length is ~4.7 signs, so this is an approximation). By word-length KL divergence:

Rank	Hypothesis	Word-length KL↓
1	Sumerian	1.8911
2	Proto-Dravidian	2.1122
3	Vedic Sanskrit / Indo-Aryan	2.1123

4	Proto-Semitic	2.1123
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Table 10. Indus Script word-structure ranking (Sumerian KL=1.89 is lowest due to longer expected mean length; Dravidian, Sanskrit, Semitic cluster at KL≈2.11). Note: all KL values are high due to the mismatch between fixed 5-sign chunking and the reference profiles (mean 2.8–3.7). Definitive Indus word-structure analysis requires actual inscription boundaries from the ICIT corpus.

9.4 Cross-script structural alignment

Linear A mean inscription length: 2.85 signs. Linear B mean: 4.00 signs. Word-length KL divergence between scripts: 14.39. The large KL divergence confirms that Linear A and Linear B have structurally different administrative text formats despite sharing sign shapes. Linear A entries are shorter (more 2-sign entries) while Linear B entries are longer (4-sign syllabic words are common). This structural dissimilarity is itself evidence against the assumption that Linear A simply encodes the same administrative Greek as Linear B in the same way.

10. Discussion

10.1 What the study establishes

The Glossa Lab toolkit is technically credible: it recovers known ancient script mappings at 96.7–100% accuracy. Linear A is definitively linguistic by every entropy and distributional measure. The anti-circularity suite is the most rigorous published test of the Greek-adjacency claim for Linear A.

10.2 What the study does not establish

The Greek-adjacent phonological claim is not supported independently of circular vocabulary evidence. The Kandles phonetic fingerprint marginally favours Luwian. This does not prove Minoan is Anatolian — the margin is small and the Luwian model is minimal — but it does mean the Greek claim requires stronger independent evidence.

10.3 Implications for Indus Script

The methodology developed here — particularly the word-structure hypothesis test and the anti-circularity controls — transfers directly to the Indus Script. The Indus hypothesis engine result (Dravidian 297 vs Sanskrit 77) is on synthetic data. The ICIT corpus from Dr. Andreas Fuls (TU Berlin), when available, will enable the same full seven-experiment anti-circularity test on real Indus inscriptions.

10.4 Limitations

- Linear A corpus size (~7,400 sign tokens) is small relative to the hypothesis search space.
- Language models for Hurrian, Luwian, and Semitic are minimal character-level corpora; this may underestimate those hypotheses.
- The Kandles phonetic fingerprint depends on a phoneme-to-colour mapping that was designed for Greek-family phonology; it may not be equally sensitive to non-IE phonological contrasts.
- The vocabulary used for matching (ku-ro, ki-re-ta, sa-ra2) was derived from Linear B phonetic values — a circular dependency that Section 7 quantifies and acknowledges.
- Indus Script analysis is on a statistical corpus model, not the actual Mahadevan sign sequence.

11. Future Work

ICIT corpus. The primary bottleneck is access to the Wells-Fuls ICIT database (4,537 Indus inscribed objects, 19,616 sign occurrences). The Fuls parser in Glossa Lab already handles the +sign-sign-sign+ notation. All 12 pipelines will be run on the full corpus when available.

Cross-administrative alignment. Many Linear A administrative formulas appear in structurally identical positions to Linear B formulas. Aligning sign groups by functional context (rather than shape) could establish structural equivalences without assuming phonetic identity.

Richer language models. The Hurrian and Luwian language models are minimal. The ETCSL Sumerian corpus, Hrozný Hittite corpus, and published Hurrian text transcriptions would substantially improve the non-Greek comparisons.

NSB entropy estimators. The Miller-Madow and Chao-Shen estimators are implemented in Glossa Lab but have not yet been applied to the small per-site Linear A corpus splits. These would improve entropy estimates for the 67- and 157-token small-site corpora.

Minimum description length decipherment. The MDL-based approach (Snyder et al. 2010) is not yet implemented. It would provide a principled Bayesian alternative to the hill-climbing engine for multi-script comparisons.

12. Conclusion

We have presented Glossa Lab, a validated computational toolkit for ancient script analysis, and applied it to Linear A under the most rigorous anti-circularity conditions yet published. The toolkit is technically credible (Linear B 100%, Ugaritic 96.7%). Linear A is definitively a linguistic script. The Greek-adjacent phonological claim does not survive when vocabulary evidence is removed. The vocabulary-independent Kandles phonetic fingerprint marginally favours Luwian/Anatolian.

Two new assumption-free methods — distributional sign clustering (Ventris grid operationalisation) and word-structure hypothesis testing — are introduced and applied to both Linear A and the Indus Script. These methods provide a path toward phoneme discovery that does not inherit the circularities of existing approaches.

The correct current claim is: *Linear A is linguistic; its word-structural profile marginally resembles Anatolian languages more than Mycenaean Greek or Semitic; and the Dravidian structural hypothesis remains the strongest available for the Indus Script. Neither claim constitutes a decipherment.*

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Appendix A: Supplementary Tables

A1. Site-by-site hypothesis ranking (full scoring, real tablet data)

Site	N	Greek	Hurrian	Luwian	Semitic	Winner	Margin
ALL	5379	56.9	17.0	17.0	17.0	greek	39.9
HT	3328	56.9	16.9	16.8	16.9	greek	40.0
KH	480	23.5	15.7	15.6	15.7	greek	7.8
ZA	673	25.9	16.9	16.9	16.8	greek	8.9
PH	272	20.4	11.9	11.7	11.8	greek	8.5
KN	158	18.5	9.7	9.7	9.8	greek	8.7
ARKH	157	8.6	9.7	9.7	9.8	semitic	0.1
MA	67	7.5	8.0	8.1	7.9	luwian	0.1
TY	77	7.6	8.2	8.0	8.1	hurrian	0.1

Green rows: large corpus sites (n≥200).

A2. Mapping ablation results (no-vocab scoring, 30 trials per level)

N signs	Greek mean	95% CI	Hurrian mean	Greek rank-1
10	16.878	[16.868,16.890]	16.942	13%
20	16.896	[16.881,16.910]	16.899	30%
30	16.899	[16.881,16.918]	16.818	57%
40	16.891	[16.871,16.910]	16.760	83%
104	16.903	[16.903,16.903]	16.966	0%

A3. Random/permutated mapping null distribution

Control	Trials	Null mean	Real	p	z	Pct>real
frequency matched random	30	16.889	16.903	0.4000	0.292	40%
cv structure preserving	30	16.891	16.903	0.4000	0.269	40%
permuted lb correspondences	30	16.889	16.903	0.4000	0.292	40%